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Refined Gabor Feature Extraction Combined with Optimized Bayesian Classification for Facial Expression Analysis

K. Nirmala Devi^{1,*}, S. Thanga Prasath² and Shanmugan Joghee³

¹Associate Professor, Department of CSE-Data Science, Madanapalle Institute of Technology & Science (MITS) - Deemed to be University, Madanapalle, Andhra Pradesh - 517325, India.

²Assistant Professor, Department of Artificial Intelligence, Madanapalle Institute of Technology & Science (MITS), Deemed to be University, Madanapalle, Andhra Pradesh - 517325, India.

³Professor & Discipline leader - Marketing, School of Business, Horizon University College, Al Jurf, Ajman, United Arab Emirates.

*Corresponding Author: K. Nirmala Devi. Email: nirmaladevik.cse@gmail.com

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ABSTRACT: Face recognition is a subdomain of computational biometrics, which is an automated technique for verifying the identity of a human based on some parameters, like the pattern in the iris and face dimensions. It is a multi-faceted process as there are only a limited images available for the purpose of training, and different images are obtained through various conditions and surroundings. There is a constant challenge that researchers face, which is how to efficiently handle the sensitivity of the classifier under illumination, expressions, and performance under low-resolution images. Conventional methods of face recognition have a lower accuracy in the classification, and hence, this paper proposes a Modified Gabor Filtering technique based on NB classifier (MGF+ENBA) for overcoming the issue. The MGF technique extracts only the most important features from the available data efficiently. The ENBA algorithm is then used for the classification of facial expression in a more accurate way, with the emotion detection, such as sad, happy, angry, fear, and neutral emotional features. The experiments were performed using the Cohn-Kanade data and the JAFFE database. The results prove that the proposed technique performs better than the other methods in terms of precision, recall, and F-measure, which proves improved accuracy.

KEYWORDS: Facial expression, Modified Gabor Filter with Enhanced Naïve Bayes Algorithm (MGF+ENBA), image processing.

1 Introduction

Facial expression detection systems are largely being researched in the AI domain. A lot of facial expression recognition (FER) applications apply traditional ML methods for the feature extraction, and these traditional methods have an overhead of poor accuracy when new images are tested. Many Artificial Intelligence-based systems are being developed for the detection of facial expressions for improving business domains, such as in customer attention marketing, health monitoring, and other intelligent systems based on emotions. Owing to a great demand and lack of supply, the facial



recognition systems with emotional intelligence have attracted a large volume of research community towards the computer vision domain [1-2].

Human emotions often facilitate more meaning than words. Thus, the importance of understanding emotions is crucial in the context of social communications. Many techniques are available for the detection of emotions, such as voice recognition, NLP, and complex techniques, including EEGs. However, a much easier and stronger way for emotion detection is through facial expressions. The common facial expressions include anger, happiness, fear, surprise, and disgust, and more complex expressions which have the combination of basic emotions are still under investigation [3-4]. Hence, a utility that can detect the motions from facial expressions is in great demand.

The conventional method has used feature selection based on hand-crafted techniques. These include techniques such as the Local Binary Patterns (LBP), the non-negative factorization (NMF), and spatial learning [5-6]. In the year 2013, various competitions on FER [21] and emotion detection were conducted, which have largely collected enough training data in real-time [7-8]. Owing to the great advancements in chip-level technologies and well-architected networks, various research has commenced in the usage of deep learning methods that have achieved a lot in the accurate detection of emotions [9-10]. Also, given the effectiveness of the training of data, deep learning has shown a dramatic efficiency for handling the most challenging issues in emotional detection.

In any image processing technique, feature selection plays a vital role in the classification part. If there are no sufficient features, the performance of the classifier will be poor. In most cases of facial expression classification, a large number of features will be generated, and only the significant features will be needed for the classification, which is done through optimization techniques [11-12]. Among the famous techniques, the Gabor Wavelet Feature extraction has attracted many researchers owing to its performance. Although widely adopted, the computational cost is very high for the coevolution of facial images with multi-bank filters for extracting the scalar and orientation parameters. Further, this technique suffers from a couple of major hindrances, such as the bandwidth of the GF is always limited to a single octave, and the filters are not optimized. These require the intervention of a log form in GF filters for efficient feature extraction.

In order to overcome this hindrance, this paper proposes a modified Gabor Filtering technique based on an NB classifier (MGF+ENBA) for overcoming the issue. The MGF technique extracts only the most important features from the available data efficiently. The ENBA algorithm is then used for the classification of facial expression in a more accurate way, with the emotion detection, such as sad, happy, angry, fear, and neutral emotional features. The rest of the paper is organized in the following manner. Section 2 provides the gist of important works in the literature on emotion detection, Section 3 explains the proposed method in detail, Section 4 elaborates on the results and discussion, and Section 5 concludes the research.

2 Related Works

Kim et al [15] gave special attention to the utilization of contrast representation, which can embed a distinct feature factor for a discriminative intent. The contrast representation that embeds a deep neural network is used for the comparison of the given image with that of a reference. The authors attempted to use a generic reference that is estimated from the given image. As a consequence, the DNNs are deployed, which can combine to form an iterative model that has an end-to-end training. In the proposed network, an attempt to disintegrate the expressive factor in a couple of steps, which includes the learning development framework, is carried out. The experiments were carried out using public data sets available, and it is seen that the proposed method has high detection accuracy.

Dewi et al [16] performed a critical review on the introduction of facial emotions and developed the LGBP-TOP combined with SVM, which tends to focus on the images of faces and lips alone. A novel algorithm was developed for extracting the features of lips and eyes using the 3D blocks, which is then used as the input for the LGBP-TOP technique. Later, the results are classified by using the SVM technique. The experimental results proved an overall accuracy of 82% and found to perform better than the entire face and also specific areas such as eyes and lips.

Luo et al [17] presented the PCA method, which is based on statistical analysis that is extracted from the grey scales of the full image. The grey-scale features are highly sensitive towards the environment, and hence, a hybrid method that combines the PCA and LBP was proposed. The LBP is determined to extract the locally available grey features on the regions surrounding the mouth and eyes. The SVM methods are used for the detection of facial emotions. The experimental results prove that the proposed method can able to classify the various expressions more effectively and can achieve a higher rate of recognition than the present traditional methods.

Krestinkaya et al [18] used the simplest method for feature selection, which is also effective. The authors have proposed a straightforward technique for the recognition of emotions to reduce the issue of the inter-class mismatch among the pixels during the process of classification. The proposed method uses the application of normalizing the pixels for the removal of offsets caused due to intensity, followed by the Min-Max metrics in a nearest classifier that is competent for the outlier suppression of features, which forms the outliers. The experimental results indicate that there is a significant improvement in the performance of 18% when the Min_max classification process is tested on the JAFFE base. The proposed technique outperforms all the other existing methods that are based on template matching.

3 Proposed Methodology

The main objective of the proposed Modified Gabor Filter with NB Algorithm (MGF+ENBA) is to improve the FER performance.

3.1 Image Dataset

All the facial emotion classification sets that are used in the proposed method are from a couple of databases, the JAFFE [19] and Cohn-Kanade [20]. The former database contains 239 images of 7 basic facial expressions. These images are taken from the female models belonging to Japan. The emotions expressed in each of the pictures are tested for their subjectivity on 60 different volunteers. The sample set of images is shown in Figure 1.



Figure 1: Sample static images from the JAFFE database



Figure 2: Image sequences from the Cohn-Kanade database

3.2 Image Acquisition

The images that are used in the proposed method are static in nature. In order to capture the images of various expressions, a Panasonic model of camera with a focal length of 7mm is used. The JPEG format is used for images with 24-bit coloring. The resolution was fixed as 4320*3240 pixels. The maximum distance between the camera and the person was kept at 4 feet, and the images of seven basic expressions of each of the persons were taken. The image sequence is shown in Figure 2.

3.3 Image Preprocessing

The pre-processing is an important step in the image processing and not an exception in the case of facial expression detection. Pre-processing aims to pick out the images that have normal intensity with uniform shapes and size, and represent only the face that has emotion. The pre-processing helps in reducing the effect of illumination and as well as lighting. The representation of the expression can be difficult to translate and rotate. To overcome these hindrances of the pointless changes, the facial images are geometrically processed before entering classification. The pre-processing technique becomes more important in the FER task. The objective of pre-processing is to gather images that have normal intensity and uniform size and shape, and depict only the face expressions.

The preprocessing is also expected to remove the effect of lighting. The area pertaining to the face of a given image is detected using the Viola-Jones method using the AdaBoost algorithm. The method uses an object detection technique that provides a comprehensive detection of objects in real time. It is initially designed for the issue of face detection. The significant features that are used by the viola are obtained from the pixels that are selected from a rectangular area, which are imposed over the picture, and show a high level of sensitivity for the vertical and horizontal lines. AdaBoost has an adaptive learning mechanism that can be used in conjunction with all other learning methods for performance improvement. AdaBoost is considered adaptive as the subsequent classifiers are built iteratively and are made to fix the instances that are misclassified by the former classifiers. At every iteration, a distributed weight is updated in such a way that the weights of every incorrect classification are increased, which enables the new classification to be focused.

The last step in the pre-processing phase is the detection of a facial picture that has emotion within the maximum intensity level of a given emotion. The mutual information-based method is used for face detection. For every frame, this information present in-between the present frame and the initial one is computed, and the particular frame that has the mutual kind of information is then selected as the one

that represents a particular emotion with the maximum level of intensity. At last, the images are brought back to the initial size. Figure 4 shows images after the preprocessing.



Figure 3: Images after the pre-processing step

3.4 Feature extraction using Modified Gabor Filters (MGF)

The Gabor Filters (GF) are generally considered as one of the best choices for getting local frequency information. However, they commonly suffer from a couple of important limitations. The total bandwidth in the case of a GF is limited to a single octave, and also, these filters are not optimal if one of them seeks spectral information that has more localization. Log-based GFs help to overcome this limitation. The DC component is nullified here and can be built with an approximate bandwidth that can be able to produce the filter that has minimal spatial extent. Log-based filters in the frequency issue are defined in terms of polar bi-coordinates using $H(f, \theta) = H_f \times H_\theta$, where H_f depicts the radial item and H_θ , the angular component.

$$H(f, \theta) = \exp \left\{ \frac{-[\ln(\frac{f}{f_0})]^2}{2[\ln(\frac{\sigma_f}{f_0})]^2} \right\} \exp \left\{ \frac{-(\theta - \theta_0)^2}{2\sigma_\theta^2} \right\} \quad (1)$$

Here, the f_0 is the frequency at the center of the filter, and θ_0 represents the direction of the filter. Σf is the constant value that defines the bandwidth (B) of the octaves. The coefficients of the MGF are then calculated in an off-line speedup. The GF filters are with different orientations and are employed through the coefficients that are calculated respectively. This gives a variety of filters.



Figure 4: Eye images used for training and testing



Figure 5: Mouth images used for training and testing

3.5 FER classification using Enhanced Naïve Bayes Algorithm (ENBA)

The NB classifier is based on the probability that has displayed effectiveness in many problems of classification [21]. The NB classifiers are simple but are effective in classification, which is based on Bayes' theorem. The Naive Bayes theorem is trained by using the images that are captured in a supervised learning environment. It is assumed that the presence of a given feature in a class is not related to the presence of any other feature. If c represents the class variable, then the features are represented using the set $\{f_1, f_2, \dots, f_k\}$, and the decision of the classifier is made by using the formula.

$$C = \arg \max_c \{p(c) \prod_{j=1}^k p(f_j|c)\} \quad (2)$$

Where the $p(c)$ represents the total samples in the set $\{c\}$ / the total no of samples is the conditional table that is learnt in the example, and k represents the feature vector length. Although there is an existence of independent assumptions, the NB has proven to be a good choice for classification for real-time data, in par with many complex classifiers. The overall architecture of the proposed scheme is given in Fig. 6.

4 Experimental Result

The JAFFE repository has a total of 596 image sequences ranging with 124 subjects in the age range between 18-35. The DB includes faces that show the basic 7 emotions for which there were $\{45, 98, 56, 58, 78, 35, 142\}$ images, respectively. The proposed method is compared with that of the Nearest Neighbor Classification and SVM for evaluating the JAFFE database along the MMSNN algorithm. The performance metrics chosen are precision, recall, and F-Measure as measures of accuracy.

4.1 Precision

The precision can be viewed as the computational parameter of accuracy or a quality, and is calculated using the formula,

$$Precision = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} \quad (3)$$

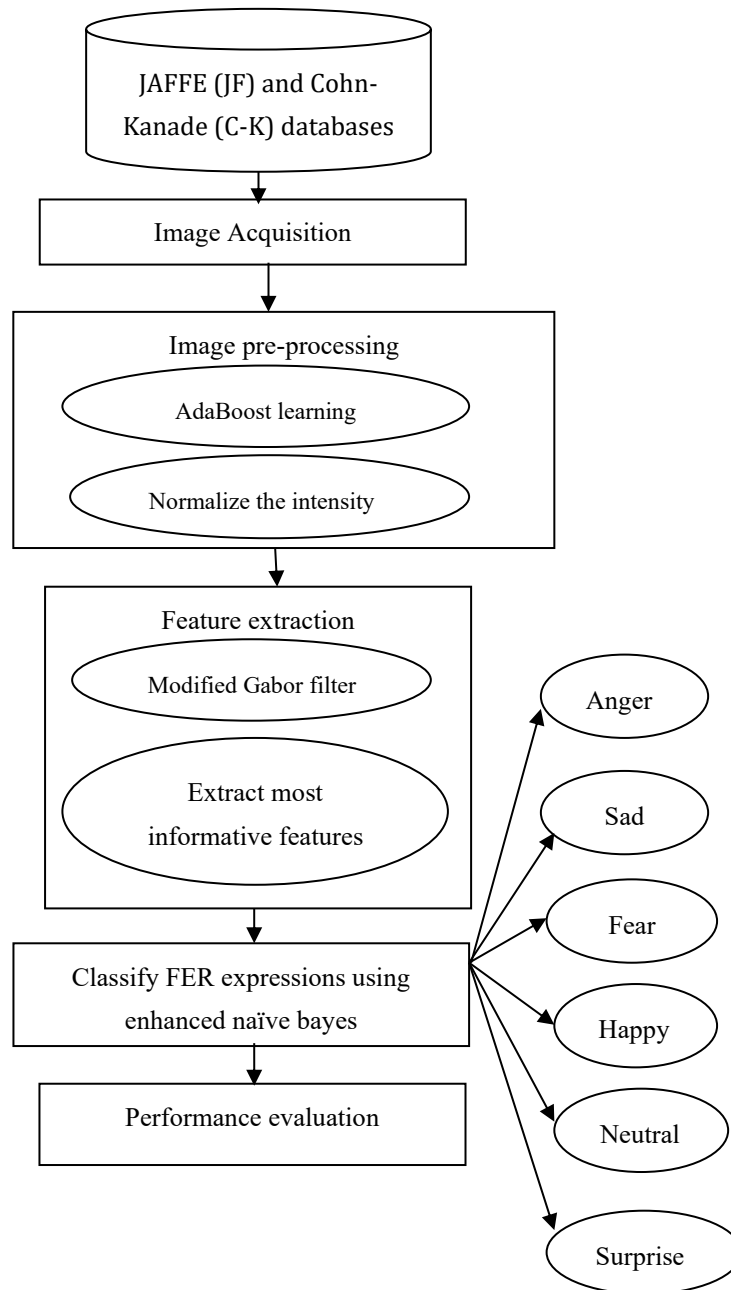


Figure 6: Overall block diagram of the proposed system

The recall is the measure of completeness, or in other words, it is a quality parameter. Usually, the high value of precision indicates that the algorithm has given back more relevant results. In the classification problem, the precision of a given class is the total count of true positives divided by the total elements that belong to the positive class.

From Figure 7, it is observed that the plot is drawn taking the existing and the proposed method and compared for its precision. In the X axis, the methods are assigned, and in the Y axis, the values of precision are plotted. It is observed that the proposed method has more precision than the other popular methods, such as the HMM and SVM. This concludes that the proposed method can increase facial expressions.

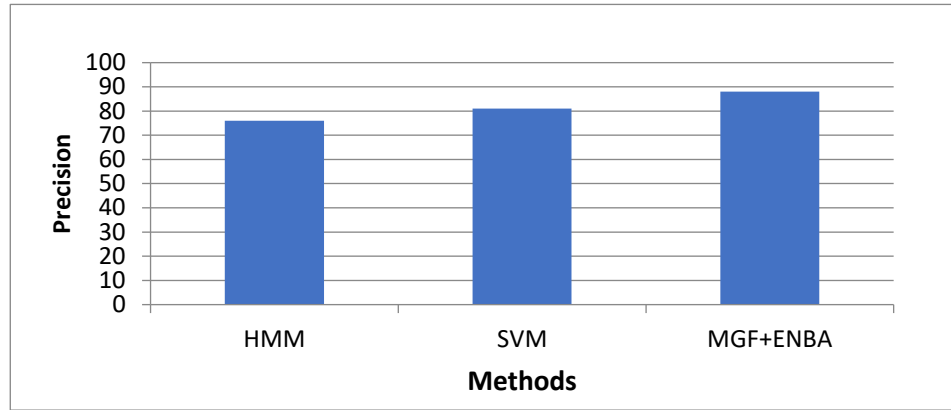


Figure 7: Precision comparison

4.2 Recall

Recall is generally defined as the no of appropriate documents received by a given search divided by the total count of existing such documents, and is calculated as,

$$Recall = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} \quad (4)$$

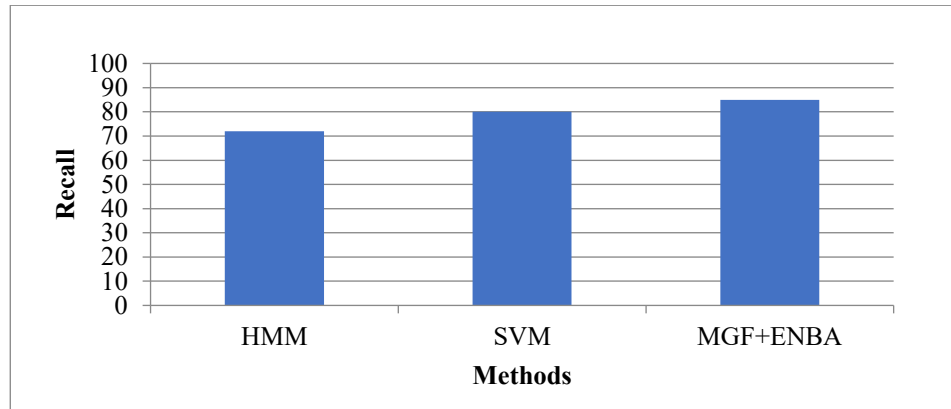


Figure 8: Recall

From Figure 8, it is seen that the comparative metric is evaluated between the proposed and the existing method. In the X axis, the methods are assigned, and in the Y axis, the values of recall are plotted. It is observed that the proposed method has more precision than the other popular methods, such as the HMM and SVM. This concludes that the proposed method MGF+ENBA can increase the detection performance of the facial expressions, such as sad, happy, anger, fear, neutral, and surprise expressions, accurately for the given databases.

4.3 F-measure

F1-score calculated as,

$$F1 - \text{score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (5)$$

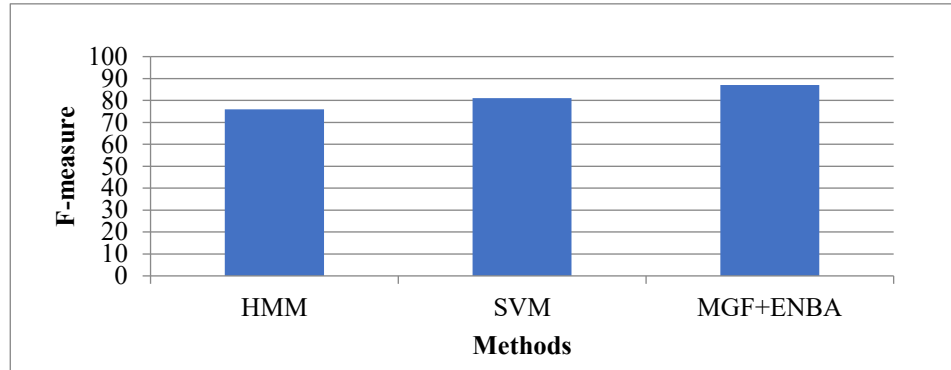


Figure 9: F-measure

From Figure 9, it is seen that the comparative metric is evaluated between the proposed and the existing method. In the X axis, the methods are assigned, and in the Y axis, the values of F-Measure are plotted. It is observed that the proposed method has more precision than the other popular methods, such as the HMM and SVM. This concludes that the proposed method MGF+ENBA can increase the detection performance of the facial expressions, such as sad, happy, anger, fear, neutral, and surprise expressions, accurately for the given databases.

4.4 Accuracy

Accuracy is the correctness of a given model, and it is calculated as the total count of actual parameters ($T_p + T_n$) of classification that are segregated by the sum of all of the classification's parameters ($T_p + T_n + F_p + F_n$). The same is computed as,

$$\text{Accuracy} = \frac{T_p + T_n}{(T_p + T_n + F_p + F_n)} \quad (6)$$

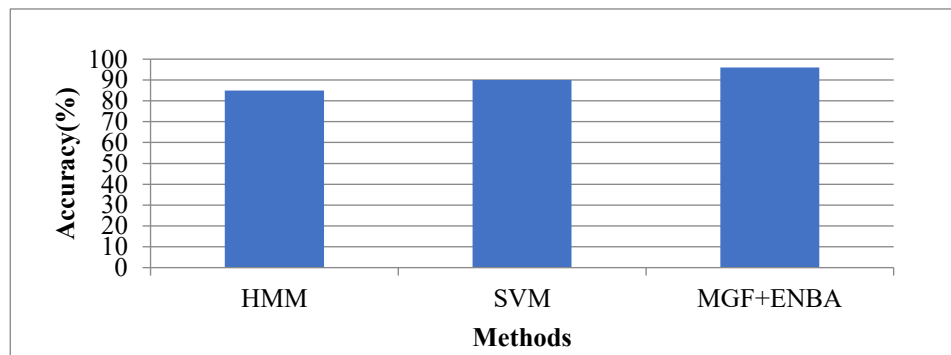


Figure 10: Accuracy

From Figure 10, it is seen that the comparative metric is evaluated between the proposed and the existing method. In the X axis, the methods are assigned, and in the Y axis, the values of accuracy are plotted. It is observed that the proposed method has more precision than the other popular methods, such as the HMM and SVM. This concludes that the proposed method MGF+ENBA can increase the detection performance of the facial expressions, such as sad, happy, anger, fear, neutral, and surprise expressions, accurately for the given databases.

5 Conclusion

In this study, an efficient method for the calculation of an accurate FER rate using the MGF+ENBA technique is presented. The FERs are based on the hybrid facial regions, which are taken for

investigation. The part of the face that shows the most discriminative information on facial emotional expression classification. The vector representing the features is extracted from the original image. Then, the significant features are selected based on the mutual information. These are classified using the ENBA classifier. The experiments are carried out using the JAFFE database images, and it is found that the proposed MGF+ENBA algorithm gives more accuracy and better performance than other competitive methods, such as HMM and SVM algorithms.

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